

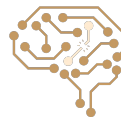
# Probability & Uncertainty



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Some of Slides have been adopted from Berkeley CS188



# So Far: Search

- Informed and Uninformed Search
- Local Search (Search Among Complete States)
- CSP (Search + Constraints)
- Adversarial Search (For Multi-agent Scenarios)



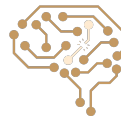
# Today: Uncertainty & Probability

- Probabilistic Reasoning!
- Forming and Updating **Beliefs** Based on **Observations**!



# Probability Recap

- Random Variables
- Joint Distributions
- Marginal Distributions
- Conditional Distribution
- Product Rule, Chain Rule and Bayes' Rule
- Inference by Enumeration



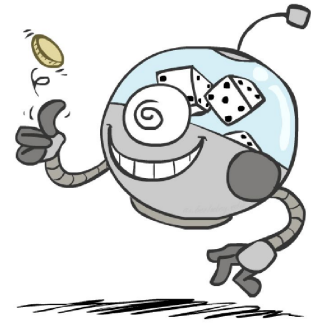
# World with Uncertainty

- **Observed variables (evidence):** Agent knows certain things about the state of the world (e.g., sensor readings or symptoms)
- **Unobserved (hidden) variables:** Agent needs to reason about other aspects (e.g. where an object is or what disease is present)
- **Model:** Agent knows something about how the known variables relate to the unknown variables
- **Probabilistic reasoning** gives us a framework for managing our beliefs and knowledge



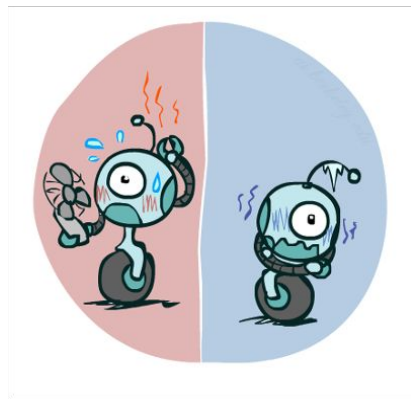
# Random Variables

- A random variable is some aspect of the world about which we (may) have uncertainty
  - $R$  = Is it raining?
  - $T$  = Is it hot or cold?
  - $D$  = How long will it take to drive to work?
- We denote random variables with capital letters
- Like variables in a CSP, random variables have domains
  - $R$  in  $\{\text{true}, \text{false}\}$  (often write as  $\{+r, -r\}$ )
  - $T$  in  $\{\text{hot}, \text{cold}\}$
  - $D$  in  $[0, +\infty)$



# Probability Distributions

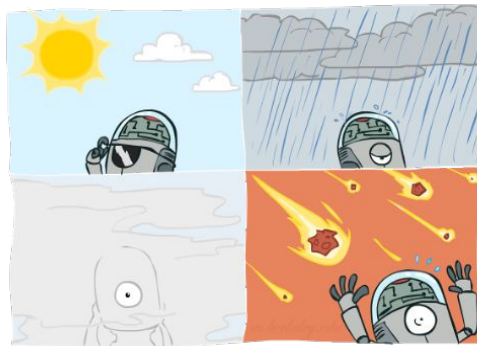
## Temperature



$P(T)$

| T    | P   |
|------|-----|
| hot  | 0.5 |
| cold | 0.5 |

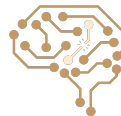
## Weather



$P(W)$

| W      | P   |
|--------|-----|
| sun    | 0.6 |
| rain   | 0.1 |
| fog    | 0.3 |
| meteor | 0.0 |

- We must have  $\forall x \ P(X = x) \geq 0$  and  $\sum_x P(X = x) = 1$





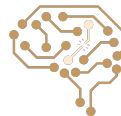
# Probability Distributions (cont.)

- Unobserved random variables have distributions
- A distribution is a TABLE of probabilities of values

| $P(T)$ |     |
|--------|-----|
| T      | P   |
| hot    | 0.5 |
| cold   | 0.5 |

| $P(W)$ |     |
|--------|-----|
| W      | P   |
| sun    | 0.6 |
| rain   | 0.1 |
| fog    | 0.3 |
| meteor | 0.0 |

- A probability (lower case value) is a single number:  $P(W = rain) = 0.1$



# Joint Distributions

- A joint distribution over a set of random variables:  $X_1, X_2, \dots, X_n$  specifies a real number for each assignment (or outcome):

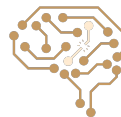
$$P(X_1 = x_1, X_2 = x_2, \dots, X_n = x_n)$$

$$P(x_1, x_2, \dots, x_n)$$

- Must obey:

$$P(x_1, x_2, \dots, x_n) \geq 0$$

$$\sum_{(x_1, x_2, \dots, x_n)} P(x_1, x_2, \dots, x_n) = 1$$



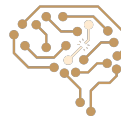
## Joint Distributions (cont.)

$$P(T, W)$$

- Example:

| T    | W    | P   |
|------|------|-----|
| hot  | sun  | 0.4 |
| hot  | rain | 0.1 |
| cold | sun  | 0.2 |
| cold | rain | 0.3 |

- Size of distribution if  $n$  variables with domain sizes  $d$ ?
- For all but the smallest distributions, impractical to write out!

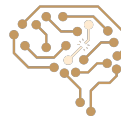


# Probabilistic Models

- A **probabilistic model** is a joint distribution over a set of random variables
- Assignments are called outcomes
- Joint Probabilities: say whether assignments (outcomes) are likely

Distribution over T,W

| T    | W    | P   |
|------|------|-----|
| hot  | sun  | 0.4 |
| hot  | rain | 0.1 |
| cold | sun  | 0.2 |
| cold | rain | 0.3 |



$$P(T, W)$$

## Events

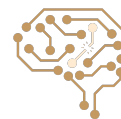
- An event is a set  $E$  of outcomes:

$$P(E) = \sum_{(x_1 \dots x_n) \in E} P(x_1 \dots x_n)$$

| T    | W    | P   |
|------|------|-----|
| hot  | sun  | 0.4 |
| hot  | rain | 0.1 |
| cold | sun  | 0.2 |
| cold | rain | 0.3 |

- Having joint distribution  $\rightarrow$  calculate the probability of any event
  - Probability that it's hot AND sunny?
  - Probability that it's hot?
  - Probability that it's hot OR sunny?

Typically, the events we care about are partial assignments, like  $P(T=\text{hot})$



# Quiz: Events

- $P(+x \text{ AND } +y)$ ?
- $P(+x)$ ?
- $P(+x \text{ OR } -y)$ ?

$$P(X, Y)$$

| X  | Y  | P   |
|----|----|-----|
| +x | +y | 0.2 |
| +x | -y | 0.3 |
| -x | +y | 0.4 |
| -x | -y | 0.1 |



# Marginal Distributions

- Marginal distributions are sub-tables which eliminate variables
- Marginalization (summing out): Combine collapsed rows by adding

| T    | W    | P   |
|------|------|-----|
| hot  | sun  | 0.4 |
| hot  | rain | 0.1 |
| cold | sun  | 0.2 |
| cold | rain | 0.3 |

$$P(t) = \sum_w P(t, w)$$


| T    | P   |
|------|-----|
| hot  | 0.5 |
| cold | 0.5 |

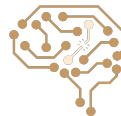
  

$$P(w) = \sum_t P(t, w)$$


| W    | P   |
|------|-----|
| sun  | 0.6 |
| rain | 0.4 |

$$P(X_1 = x_1) = \sum_{x_2} P(X_1 = x_1, X_2 = x_2)$$

 hidden (unobserved) variables



# Quiz: Marginal Distributions

$P(X, Y)$

| X  | Y  | P   |
|----|----|-----|
| +x | +y | 0.2 |
| +x | -y | 0.3 |
| -x | +y | 0.4 |
| -x | -y | 0.1 |

$$P(x) = \sum_y P(x, y)$$

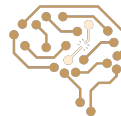

$P(X)$

| X  | P |
|----|---|
| +x |   |
| -x |   |

$$P(y) = \sum_x P(x, y)$$


$P(Y)$

| Y  | P |
|----|---|
| +y |   |
| -y |   |

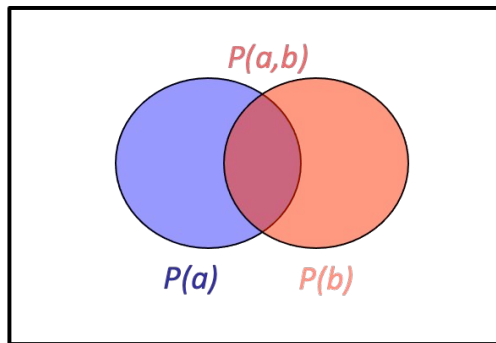




# Conditional Probabilities

- A simple relation between joint and conditional probabilities

$$P(a|b) = \frac{P(a,b)}{P(b)}$$

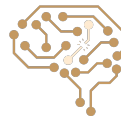


$P(T, W)$

| T    | W    | P   |
|------|------|-----|
| hot  | sun  | 0.4 |
| hot  | rain | 0.1 |
| cold | sun  | 0.2 |
| cold | rain | 0.3 |

$$P(W = s|T = c) = \frac{P(W = s, T = c)}{P(T = c)} = \frac{0.2}{0.5} = 0.4$$

$$\begin{aligned} &= P(W = s, T = c) + P(W = r, T = c) \\ &= 0.2 + 0.3 = 0.5 \end{aligned}$$



# Quiz: Conditional Probabilities

$P(X, Y)$

| X  | Y  | P   |
|----|----|-----|
| +x | +y | 0.2 |
| +x | -y | 0.3 |
| -x | +y | 0.4 |
| -x | -y | 0.1 |

$$P(a|b) = \frac{P(a, b)}{P(b)}$$

- $P(+x \mid +y) ?$
- $P(-x \mid +y) ?$
- $P(-y \mid +x) ?$



# Conditional Distributions

$$P(a|b) = \frac{P(a, b)}{P(b)}$$

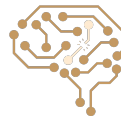
- Conditional distributions are probability distributions over some variables given fixed values of others

Conditional Distributions

|          |                        |     |
|----------|------------------------|-----|
| $P(W T)$ | $P(W T = \text{hot})$  |     |
|          | W                      | P   |
|          | sun                    | 0.8 |
|          | rain                   | 0.2 |
|          | $P(W T = \text{cold})$ |     |
|          | W                      | P   |
|          | sun                    | 0.4 |
|          | rain                   | 0.6 |

Joint Distribution

|           |      |     |
|-----------|------|-----|
| $P(T, W)$ |      |     |
| T         | W    | P   |
| hot       | sun  | 0.4 |
| hot       | rain | 0.1 |
| cold      | sun  | 0.2 |
| cold      | rain | 0.3 |



# Normalization Trick

$P(T, W)$

| T    | W    | P   |
|------|------|-----|
| hot  | sun  | 0.4 |
| hot  | rain | 0.1 |
| cold | sun  | 0.2 |
| cold | rain | 0.3 |

$$\begin{aligned}P(W = s|T = c) &= \frac{P(W = s, T = c)}{P(T = c)} \\&= \frac{P(W = s, T = c)}{P(W = s, T = c) + P(W = r, T = c)} \\&= \frac{0.2}{0.2 + 0.3} = 0.4\end{aligned}$$

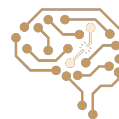


$P(W|T = c)$

| W    | P   |
|------|-----|
| sun  | 0.4 |
| rain | 0.6 |

$$\begin{aligned}P(W = r|T = c) &= \frac{P(W = r, T = c)}{P(T = c)} \\&= \frac{P(W = r, T = c)}{P(W = s, T = c) + P(W = r, T = c)} \\&= \frac{0.3}{0.2 + 0.3} = 0.6\end{aligned}$$

$$P(x_1|x_2) = \frac{P(x_1, x_2)}{P(x_2)} = \frac{P(x_1, x_2)}{\sum_{x_1} P(x_1, x_2)}$$



# Normalization Trick (cont.)

$$P(T, W)$$

| T    | W    | P   |
|------|------|-----|
| hot  | sun  | 0.4 |
| hot  | rain | 0.1 |
| cold | sun  | 0.2 |
| cold | rain | 0.3 |

**SELECT** the joint probabilities matching the evidence



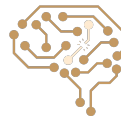
| T    | W    | P   |
|------|------|-----|
| cold | sun  | 0.2 |
| cold | rain | 0.3 |

**NORMALIZE** the selection (make it sum to one)



$$P(W|T = c)$$

| W    | P   |
|------|-----|
| sun  | 0.4 |
| rain | 0.6 |



# Quiz: Normalization Trick

■  $P(X \mid Y=-y)$  ?

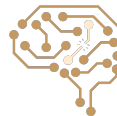
$P(X, Y)$

| X  | Y  | P   |
|----|----|-----|
| +x | +y | 0.2 |
| +x | -y | 0.3 |
| -x | +y | 0.4 |
| -x | -y | 0.1 |

**SELECT** the joint  
probabilities  
matching the  
evidence



**NORMALIZE** the  
selection  
(make it sum to one)



# Probabilistic Inference

- Probabilistic inference: compute a **desired probability** from other **known probabilities** (e.g. conditional from joint)
- We generally compute conditional probabilities
  - $P(\text{on time} \mid \text{no reported accidents}) = 0.90$
  - These represent the agent's beliefs given the evidence
- Probabilities change with new evidence:
  - $P(\text{on time} \mid \text{no accidents, 5 a.m.}) = 0.95$
  - $P(\text{on time} \mid \text{no accidents, 5 a.m., raining}) = 0.80$
  - Observing new evidence causes beliefs to be updated



# Inference by Enumeration

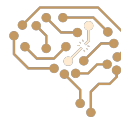
- General case:

- Evidence variables:  $E_1 \dots E_k = e_1 \dots e_k$
  - Query\* variable:  $Q$
  - Hidden variables:  $H_1 \dots H_r$
- }  $X_1, X_2, \dots, X_n$   
*All variables*

- We want:

*\* Works fine with  
multiple query  
variables, too*

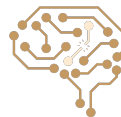
$$P(Q|e_1 \dots e_k)$$





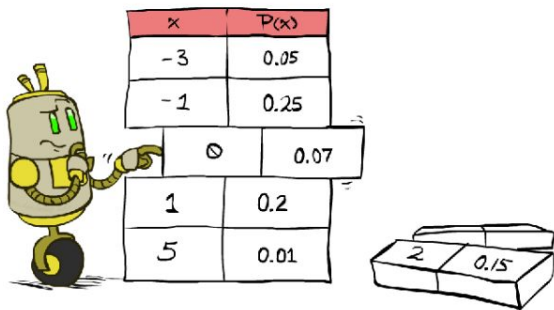
# Inference by Enumeration Example

- Example:
  - Variables: arrival time, accident status, weather
  - $P(\text{arrival time} \mid \text{no accidents})$ 
    - Evidence: accident status
    - Query: arrival time
    - Hidden: weather
  - $P(\text{arrival time} \mid \text{no accidents, rain})$ 
    - Evidence: accident status, weather
    - Query: arrival time
    - Hidden: none

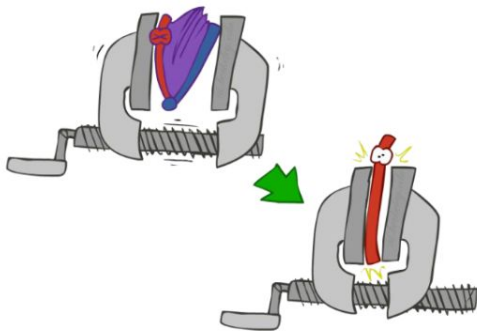


# Inference by Enumeration (cont.)

- Step 1: **Select** the entries consistent with the evidence



- Step 2: **Sum** out H to get joint of Query and evidence



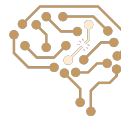
$$P(Q, e_1 \dots e_k) = \sum_{h_1 \dots h_r} \underbrace{P(Q, h_1 \dots h_r, e_1 \dots e_k)}_{X_1, X_2, \dots, X_n}$$

- Step 3: **Normalize**

$$\times \frac{1}{Z}$$

$$Z = \sum_q P(Q, e_1 \dots e_k)$$

$$P(Q|e_1 \dots e_k) = \frac{1}{Z} P(Q, e_1 \dots e_k)$$



# Quiz: Inference by Enumeration

- $P(W)?$   
↑  
query

| S      | T    | W    | P    |
|--------|------|------|------|
| summer | hot  | sun  | 0.30 |
| summer | hot  | rain | 0.05 |
| summer | cold | sun  | 0.10 |
| summer | cold | rain | 0.05 |
| winter | hot  | sun  | 0.10 |
| winter | hot  | rain | 0.05 |
| winter | cold | sun  | 0.15 |
| winter | cold | rain | 0.20 |



# Solution: Inference by Enumeration

- $P(W)$ ?

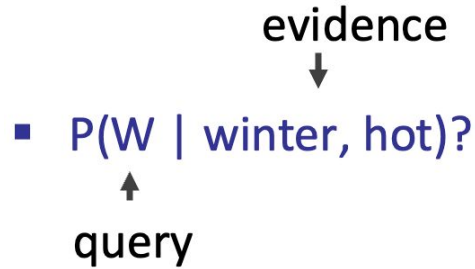
$$P(\text{sun}) = 0.3 + 0.1 + 0.1 + 0.15 = 0.65$$

$$P(\text{rain}) = 0.05 + 0.05 + 0.05 + 0.20 = 0.35$$

| S      | T    | W    | P    |
|--------|------|------|------|
| summer | hot  | sun  | 0.30 |
| summer | hot  | rain | 0.05 |
| summer | cold | sun  | 0.10 |
| summer | cold | rain | 0.05 |
| winter | hot  | sun  | 0.10 |
| winter | hot  | rain | 0.05 |
| winter | cold | sun  | 0.15 |
| winter | cold | rain | 0.20 |



# Quiz: Inference by Enumeration



| S      | T    | W    | P    |
|--------|------|------|------|
| summer | hot  | sun  | 0.30 |
| summer | hot  | rain | 0.05 |
| summer | cold | sun  | 0.10 |
| summer | cold | rain | 0.05 |
| winter | hot  | sun  | 0.10 |
| winter | hot  | rain | 0.05 |
| winter | cold | sun  | 0.15 |
| winter | cold | rain | 0.20 |



# Solution: Inference by Enumeration

- $P(W \mid \text{winter, hot})?$

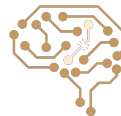
unnormalized  $P(\text{sun} \mid \text{winter, hot}) = 0.10$

unnormalized  $P(\text{rain} \mid \text{winter, hot}) = 0.05$

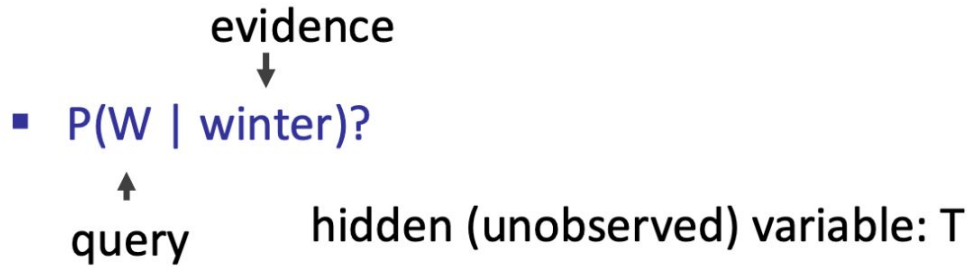
$P(\text{sun} \mid \text{winter, hot}) = 0.10 / 0.15 = 2/3$

$P(\text{rain} \mid \text{winter, hot}) = 0.05 / 0.15 = 1/3$

| S      | T    | W    | P    |
|--------|------|------|------|
| summer | hot  | sun  | 0.30 |
| summer | hot  | rain | 0.05 |
| summer | cold | sun  | 0.10 |
| summer | cold | rain | 0.05 |
| winter | hot  | sun  | 0.10 |
| winter | hot  | rain | 0.05 |
| winter | cold | sun  | 0.15 |
| winter | cold | rain | 0.20 |



# Quiz: Inference by Enumeration



| S      | T    | W    | P    |
|--------|------|------|------|
| summer | hot  | sun  | 0.30 |
| summer | hot  | rain | 0.05 |
| summer | cold | sun  | 0.10 |
| summer | cold | rain | 0.05 |
| winter | hot  | sun  | 0.10 |
| winter | hot  | rain | 0.05 |
| winter | cold | sun  | 0.15 |
| winter | cold | rain | 0.20 |



# Solution: Inference by Enumeration

- $P(W \mid \text{winter})?$

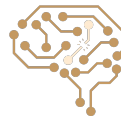
unnormalized  $P(\text{sun} \mid \text{winter}) = 0.1 + 0.15 = 0.25$

unnormalized  $P(\text{rain} \mid \text{winter}) = 0.05 + 0.20 = 0.25$

$P(\text{sun} \mid \text{winter}) = 0.25 / 0.50 = 0.5$

$P(\text{rain} \mid \text{winter}) = 0.25 / 0.50 = 0.5$

| S      | T    | W    | P    |
|--------|------|------|------|
| summer | hot  | sun  | 0.30 |
| summer | hot  | rain | 0.05 |
| summer | cold | sun  | 0.10 |
| summer | cold | rain | 0.05 |
| winter | hot  | sun  | 0.10 |
| winter | hot  | rain | 0.05 |
| winter | cold | sun  | 0.15 |
| winter | cold | rain | 0.20 |

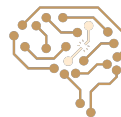
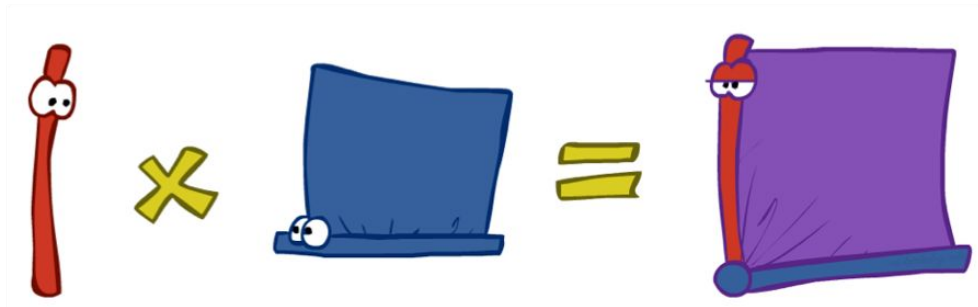




# The Product Rule

- Sometimes have conditional distributions but want the joint

$$P(y)P(x|y) = P(x, y) \quad \longleftrightarrow \quad P(x|y) = \frac{P(x, y)}{P(y)}$$



## Quiz: The Product Rule

$$P(y)P(x|y) = P(x, y)$$

- Example:

$P(W)$

| R    | P   |
|------|-----|
| sun  | 0.8 |
| rain | 0.2 |

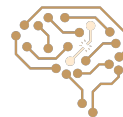
$P(D|W)$

| D   | W    | P   |
|-----|------|-----|
| wet | sun  | 0.1 |
| dry | sun  | 0.9 |
| wet | rain | 0.7 |
| dry | rain | 0.3 |



$P(D, W)$

| D   | W    | P |
|-----|------|---|
| wet | sun  |   |
| dry | sun  |   |
| wet | rain |   |
| dry | rain |   |

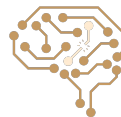


# The Chain Rule

- More generally, can always write any joint distribution as an incremental product of conditional distributions

$$P(x_1, x_2, x_3) = P(x_1)P(x_2|x_1)P(x_3|x_1, x_2)$$

$$P(x_1, x_2, \dots x_n) = \prod_i P(x_i|x_1 \dots x_{i-1})$$



# Bayes' Rule

- Two ways to factor a joint distribution over two variables:

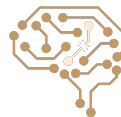
$$P(x, y) = P(x|y)P(y) = P(y|x)P(x)$$

- Dividing, we get:

$$P(x|y) = \frac{P(y|x)P(x)}{P(y)}$$

- Why is this at all helpful?
  - Lets us build one conditional from its reverse
  - Often one conditional is tricky but the other one is simple

That's my rule!

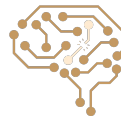


# Inference with Bayes' Rule

- Diagnostic probability from causal probability:

$$P(\text{cause}|\text{effect}) = \frac{P(\text{effect}|\text{cause})P(\text{cause})}{P(\text{effect})}$$

- Example:
  - Cause = Covid-19
  - Effect = Death



# Quiz: Inference with Bayes' Rule

- Given:

$$P(W)$$

| R    | P   |
|------|-----|
| sun  | 0.8 |
| rain | 0.2 |

$$P(D|W)$$

| D   | W    | P   |
|-----|------|-----|
| wet | sun  | 0.1 |
| dry | sun  | 0.9 |
| wet | rain | 0.7 |
| dry | rain | 0.3 |

- What is  $P(W \mid \text{dry})$  ?

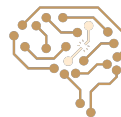


# Independence

- Two variables are independent if:

$$\forall x, y : P(x, y) = P(x)P(y)$$

- This says that their joint distribution factors into a product two simpler distributions
- Another form:  $\forall x, y : P(x|y) = P(x)$
- We write:  $X \perp\!\!\!\perp Y$



# Quiz: Independance

$P_1(T, W)$

| T    | W    | P   |
|------|------|-----|
| hot  | sun  | 0.4 |
| hot  | rain | 0.1 |
| cold | sun  | 0.2 |
| cold | rain | 0.3 |

$P(T)$

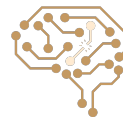
| T    | P   |
|------|-----|
| hot  | 0.5 |
| cold | 0.5 |

$P_2(T, W)$

| T    | W    | P   |
|------|------|-----|
| hot  | sun  | 0.3 |
| hot  | rain | 0.2 |
| cold | sun  | 0.3 |
| cold | rain | 0.2 |

$P(W)$

| W    | P   |
|------|-----|
| sun  | 0.6 |
| rain | 0.4 |





# Conditional Independence

- X is conditionally independent of Y given Z

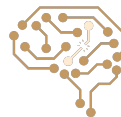
$$X \perp\!\!\!\perp Y | Z$$

- if and only if:

$$\forall x, y, z : P(x, y | z) = P(x | z) P(y | z)$$

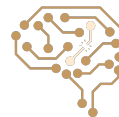
- or, equivalently, if and only if

$$\forall x, y, z : P(x | z, y) = P(x | z)$$



# Conditional Independence

- Variables:
  - Traffic
  - Umbrella
  - Raining



# Conditional Independence and the Chain Rule

- Chain rule:

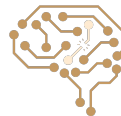
$$P(X_1, X_2, \dots, X_n) = P(X_1)P(X_2|X_1)P(X_3|X_1, X_2) \dots$$

- Trivial decomposition:

$$P(\text{Traffic, Rain, Umbrella}) = \\ P(\text{Rain})P(\text{Traffic}|\text{Rain})P(\text{Umbrella}|\text{Rain, Traffic})$$

- With assumption of conditional independence:

$$P(\text{Traffic, Rain, Umbrella}) = \\ P(\text{Rain})P(\text{Traffic}|\text{Rain})P(\text{Umbrella}|\text{Rain})$$



# Summary

- **Conditional probability**  $P(x|y) = \frac{P(x, y)}{P(y)}$
- **Product rule**  $P(x, y) = P(x|y)P(y)$
- **Chain rule** 
$$\begin{aligned} P(X_1, X_2, \dots, X_n) &= P(X_1)P(X_2|X_1)P(X_3|X_1, X_2) \dots \\ &= \prod_{i=1}^n P(X_i|X_1, \dots, X_{i-1}) \end{aligned}$$
- **X, Y independent if and only if:**  $\forall x, y : P(x, y) = P(x)P(y)$
- **X and Y are conditionally independent given Z if and only if:**  $X \perp\!\!\!\perp Y | Z$   
$$\forall x, y, z : P(x, y|z) = P(x|z)P(y|z)$$



# Next Session: Bayesian Networks

